

Exception Handling

Q315

```
class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int x,a=12,b=0;
        x=a/b; //
        System.out.println("End");
    }
}
```

////////////////////////////////////
////////////////

Q316

```
class Demo{  
    public static void main(String args[]){  
        System.out.println("Start");  
        double x,a=12,b=0;  
        x=a/b; //  
        System.out.println(x);  
        System.out.println("End");  
    }  
}
```

////////////////////////////////////
////////////////

Q317

```
class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int[] ar={100,200,300};
        System.out.println(ar[0]);
        System.out.println(ar[1]);
        System.out.println(ar[2]);
        System.out.println(ar[3]); //?
        System.out.println("End");
    }
}
```

```
////////////////////////////////////  
////////////////////////////////////
```

Q318

```
class Demo{  
    public static void main(String args[]){  
        System.out.println("Start");  
        String s1="abcd";  
        System.out.println("char at 4 : "+s1.charAt(4));  
        System.out.println("End");  
    }  
}
```

```
////////////////////////////////////  
////////////////////////////////////
```

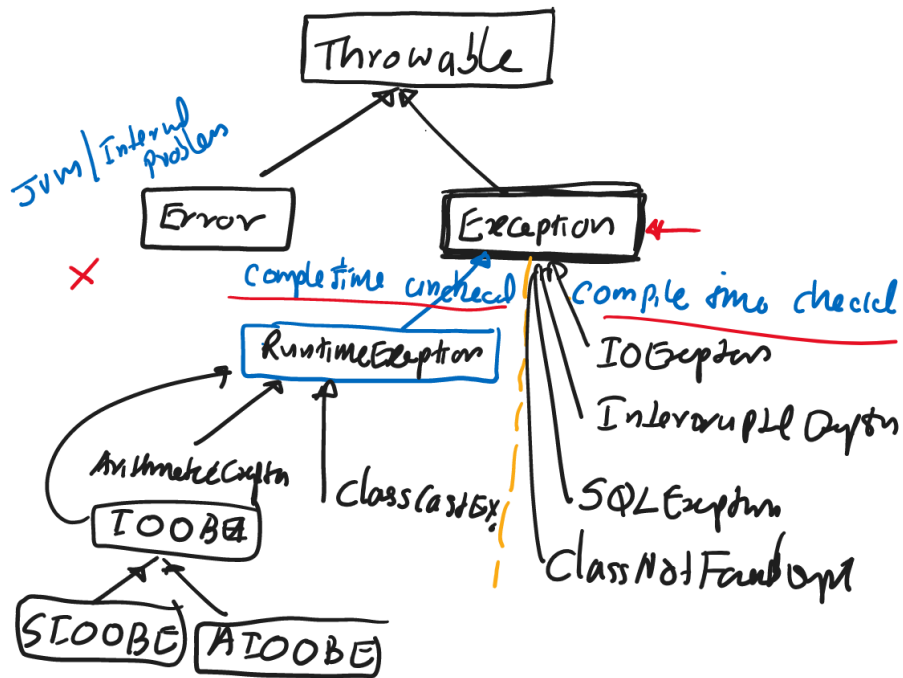
Q319 Using “try/catch” block

=====

```
class Demo{  
    public static void main(String args[]){  
        System.out.println("Start");  
        int x,a=12,b=0;  
        try{  
            x=a/b; //  
        }catch(ArithmeticException ex){  
            x=-1;  
        }  
        System.out.println("x : "+x);  
        System.out.println("End");  
    }  
}
```

```
////////////////////////////////////  
////////////////////////////////////
```

Q320



////////////////////////////////////
 //////////////////////////////////

Q321 Case I

=====

```

class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int x,a=12,b=0;
        try{
            x=a/b; //
            System.out.println("Inside try");
        }catch(ArithmeticException ex){
            x=-1;
        }
        System.out.println("x : "+x);
        System.out.println("End");
    }
}
  
```



```

class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int[] ar={10,20,30};
        try{
            ar[3]=100;
            System.out.println("Inside try");
        }catch(RuntimeException ex/*nser code here*/){
            System.out.println("Inside catch block");
        }
        System.out.println("End");
    }
}

```

//A. ArrayIndexOutOfBoundsException ex

//B. IndexOutOfBoundsException ex

//B. RuntimeException ex

//C. Exception ex

//C. Throwable ex

```

////////////////////////////////////
////////////////////////////////////

```

Q325 Case V

=====

```

class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int[] ar={10,20,30};
        try{
            ar[3]=100;
            System.out.println("Inside try");
        }catch(RuntimeException ex){
            System.out.println(ex.getMessage());
        }
        System.out.println("End");
    }
}

```

```

////////////////////////////////////
////////////////////////////////////

```

Q326 Case VI

=====

```

class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int[] ar={10,20,30};
    }
}

```



```
//-----
class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int[] ar={10,20,30};
        int x;
        try{
            //
            x=100;
        }catch(RuntimeException ex){
            x=100;
        }
        System.out.println(x);
    }
}
```

```
//-----
class Demo{
    public static void main(String args[]){
        System.out.println("Start");
        int[] ar={10,20,30};
        final int x;
        try{
            x=100;
        }catch(RuntimeException ex){
            x=100;
        }
        System.out.println(x);
    }
}
```

```
////////////////////////////////////
////////////////////////////////////
```

Nested try/catch

```
////////////////////////////////////
////////////////////////////////////
```

Q329

```
class Demo{
    public static void main(String args[]){
        System.out.println("Start");

        int[] ar={10,20,30};
        int x,a=4,b=0;
        try{
            try{
```



```
////////////////////  
////////////////
```

=====

```
////////////////////  
//////////
```

=====

```
class Demo{
    public static void main(String args[]){
        System.out.println("Start");
    }
}
```

```

        int[] ar={10,20,30};
        String s="abcd";
        int x=5,a=2;
        try{
            ar[x]=s.charAt(0)/a;
        }catch(ArithmeticException ex){
            System.out.println("Arithmetic : "+ex.getMessage());
        }
        ;
        catch(ArrayIndexOutOfBoundsException ex){
            System.out.println("Array Index : "+ex.getMessage());
        }catch(StringIndexOutOfBoundsException ex){
            System.out.println("String Index : "+ex.getMessage());
        }
        System.out.println("End.");
    }
}

```

```

////////////////////////////////////
////////////////////////////////////

```

Q334 Case II for Option 2

=====

```

class Demo{
    public static void main(String args[]){
        System.out.println("Start");

        int[] ar={10,20,30};
        String s="abcd";
        int x=5,a=2;
        try{
            ar[x]=s.charAt(0)/a;
        }catch(ArithmeticException ex){
            System.out.println("Arithmetic : "+ex.getMessage());
        }catch(ArrayIndexOutOfBoundsException ex){
            System.out.println("Array Index : "+ex.getMessage());
        }catch(IndexOutOfBoundsException ex){
            System.out.println("Any Index : "+ex.getMessage());
        }catch(StringIndexOutOfBoundsException ex){ //
            System.out.println("String Index : "+ex.getMessage());
        }
        System.out.println("End.");
    }
}

```

```
}
```

```
////////////////////////////////////  
////////////////////////////////////
```

Q335 Case 3 for Option 2

=====

```
class Demo{  
    public static void main(String args[]){  
        System.out.println("Start");  
  
        int[] ar={10,20,30};  
        String s="abcd";  
        int x=5,a=2;  
        try{  
            ar[x]=s.charAt(0)/a;  
        }catch(ArithmeticException ex){  
            System.out.println("Arithmetic : "+ex.getMessage());  
        }catch(ArrayIndexOutOfBoundsException ex){  
            System.out.println("Array Index : "+ex.getMessage());  
        }catch(StringIndexOutOfBoundsException ex){ //  
            System.out.println("String Index : "+ex.getMessage());  
        }catch(IndexOutOfBoundsException ex){  
            System.out.println("Any Index : "+ex.getMessage());  
        }catch(RuntimeException ex){  
            System.out.println("Any Runtime : "+ex.getMessage());  
        }catch(Exception ex){  
            System.out.println("Any Exception : "+ex.getMessage());  
        }  
        System.out.println("End.");  
    }  
}
```

```
////////////////////////////////////  
////////////////////////////////////
```

Q336 Case IV for Option 2

=====

```
class Demo{  
    public static void main(String args[]){  
        System.out.println("Start");  
  
        int[] ar={10,20,30};  
        String s="abcd";  
        int x=5,a=2;  
        try{
```

```

        ar[x]=s.charAt(0)/a;
    }catch(ArithmeticException ex){
        System.out.println("Arithmetic : "+ex.getMessage());
    }catch(ArrayIndexOutOfBoundsException | StringIndexOutOfBoundsException ex){
        System.out.println("Index : "+ex.getMessage());
    }catch(IndexOutOfBoundsException ex){
        System.out.println("Any Index : "+ex.getMessage());
    }catch(RuntimeException ex){
        System.out.println("Any Runtime : "+ex.getMessage());
    }
    System.out.println("End.");
}
}

```

//////////////////////////////////////
 //////////////////////////////////

finally block

//////////////////////////////////////
 //////////////////////////////////

Q337

```

class Demo{
    public static void main(String args[]){
        System.out.println("Start");

        int[] ar={10,20,30};
        String s="abcd";
        int x=5,a=2;
        try{
            //ar[x]=s.charAt(0)/a;
        }catch(ArithmeticException ex){
            System.out.println("Arithmetic : "+ex.getMessage());
        }catch(ArrayIndexOutOfBoundsException | StringIndexOutOfBoundsException ex){
            System.out.println("Index : "+ex.getMessage());
        }catch(IndexOutOfBoundsException ex){
            System.out.println("Any Index : "+ex.getMessage());
        }catch(RuntimeException ex){
            System.out.println("Any Runtime : "+ex.getMessage());
        }finally{
            System.out.println("finally block..");
        }
    }
}

```

```

        System.out.println("End.");
    }
}

////////////////////////////////////
////////////////////////////////////
Q338
class Demo{
    public static void main(String args[]){
        System.out.println("Start");

        int[] ar={10,20,30};
        String s="abcd";
        int x=5,a=2;
        try{
            ar[x]=s.charAt(0)/a;
        }catch(ArithmeticException ex){
            System.out.println("Arithmetic : "+ex.getMessage());
        }catch(ArrayIndexOutOfBoundsException | StringIndexOutOfBoundsException ex){
            System.out.println("Index : "+ex.getMessage());
        }catch(IndexOutOfBoundsException ex){
            System.out.println("Any Index : "+ex.getMessage());
        }catch(RuntimeException ex){
            System.out.println("Any Runtime : "+ex.getMessage());
        }finally{
            System.out.println("finally block..");
        }
        System.out.println("End.");
    }
}

////////////////////////////////////
////////////////////////////////////

```

```

Q339
class Demo{
    public static void main(String args[]){
        System.out.println("Start");

        int[] ar={10,20,30};
        String s="abcd";
        int x=5,a=2;
        try{
            x=45/0;
        }
    }
}

```

[illegible][illegible]

```
class Demo{
    public static boolean isAdded(){
        //
        //
    }
}
```

////////////////////////////////////
////////////////

////////////////////////////////////
////////////////

```
////////////////////////////////////  
////////////////
```

[illegible]

Q344

```
class Demo{
    public static int toInt(String sNum){
        char[] chArray=sNum.toCharArray();
        int num=0;
        for (int i = 0; i < chArray.length; i++){
            num*=10;
            switch(chArray[i]){
                //case '0':num+=0;break;
                case '1':num+=1;break;
                case '2':num+=2;break;
                case '3':num+=3;break;
                case '4':num+=4;break;
                case '5':num+=5;break;
                case '6':num+=6;break;
                case '7':num+=7;break;
                case '8':num+=8;break;
                case '9':num+=9;break;
            }
        }
        return num;
    }
    public static void main(String args[]){
        String sNum="100";
        int num=toInt(sNum);
        System.out.println(++num); //101
        System.out.println(toInt("12345"));
        System.out.println(toInt("12000"));
        System.out.println(toInt("0"));
    }
}
```

```
////////////////////////////////////
////////////////////////////////////
```

Q345 From Q344

=====

```
class Demo{
    public static int toInt(String sNum){
        char[] chArray=sNum.toCharArray();
        int num=0;
        for (int i = 0; i < chArray.length; i++){
            num*=10;
            num+=Character.getNumericValue(chArray[i]);
        }
    }
}
```

```
class Demo{
    public static void myMethod()throws ArithmeticException, ClassCastException,
    NumberFormatException{

    }
    public static void main(String args[]){

    }
}
```

Compile time checked and Compile Unchecked Exceptions

//
//

Q348

```
class Demo{
    public static void main(String args[]){
        Thread.sleep(100);
    }
}
```

//
//

Q349

```
/*
class Thread{
    public static void sleep(long x)throws InterruptedException{

    }
*/
class Demo{
    public static void main(String args[]){
        try{
            Thread.sleep(100);
        }catch(InterruptedException ex){

        }
    }
}
```

//
//

Q350

```
import java.io.*;
class FileReader{
    public static void read()throws IOException{

    }
}
```

```
        public static void write()throws RuntimeException{  
  
    }  
}  
class Demo{  
    public static void main(String args[]){  
        FileReader.write(); //Legal, RuntimeException-->Compile time unchecked  
        FileReader.read(); //IOException-->Compiletime checked  
    }  
}  
  
/////////////////////////////////////  
/////////////////////////  
Q351  
import java.io.*;  
class CustomerController{  
    public static void addCustomer(String customer)throws IOException{  
        FileWriter fr=new FileWriter("Customer.txt");  
        fr.write(customer);  
        fr.flush();  
    }  
}  
class Demo{  
    public static void main(String args[]){  
        try{  
            CustomerController.addCustomer("C001-Niroth");  
        }catch(IOException ex){  
  
        }  
    }  
}  
  
/////////////////////////////////////  
/////////////////////////  
Q352  
class NoElementException extends RuntimeException{  
    public String getMessage(){  
        return "there is no second max";  
    }  
}  
class Demo{  
    public static int secondMax(int[] ar)throws NoElementException{  
        int max=Integer.MIN_VALUE;  
        int secondMax=Integer.MIN_VALUE;  
        for (int i = 0; i < ar.length; i++){
```

```

        if(ar[i]>max){
            secondMax=max;
            max=ar[i];
        }else if(ar[i]>secondMax && ar[i]!=max){
            secondMax=ar[i];
        }
    }
    if(secondMax==Integer.MIN_VALUE){
        throw new NoSuchElementException();
    }else{
        return secondMax;
    }
}

public static void main(String args[]){
    int[] ar={12,12,12,12,12};
    int[] br={10,12,10,12,12};
    int[] cr={10,20,30,40,50};

    try{
        System.out.println("second max of br : "+secondMax(br)); //12
        System.out.println("second max of cr : "+secondMax(cr)); //40
        System.out.println("second max of ar : "+secondMax(ar)); //throws an
exception
    }catch(NoSuchElementException ex){
        System.out.println(ex.getMessage());
    }
}
}

```

```

////////////////////////////////////
////////////////////////////////////

```

Q353

```

class MyException1 extends Exception{ //Compile time checked

}

class MyException2 extends RuntimeException{ //Compile time unchecked

}

```